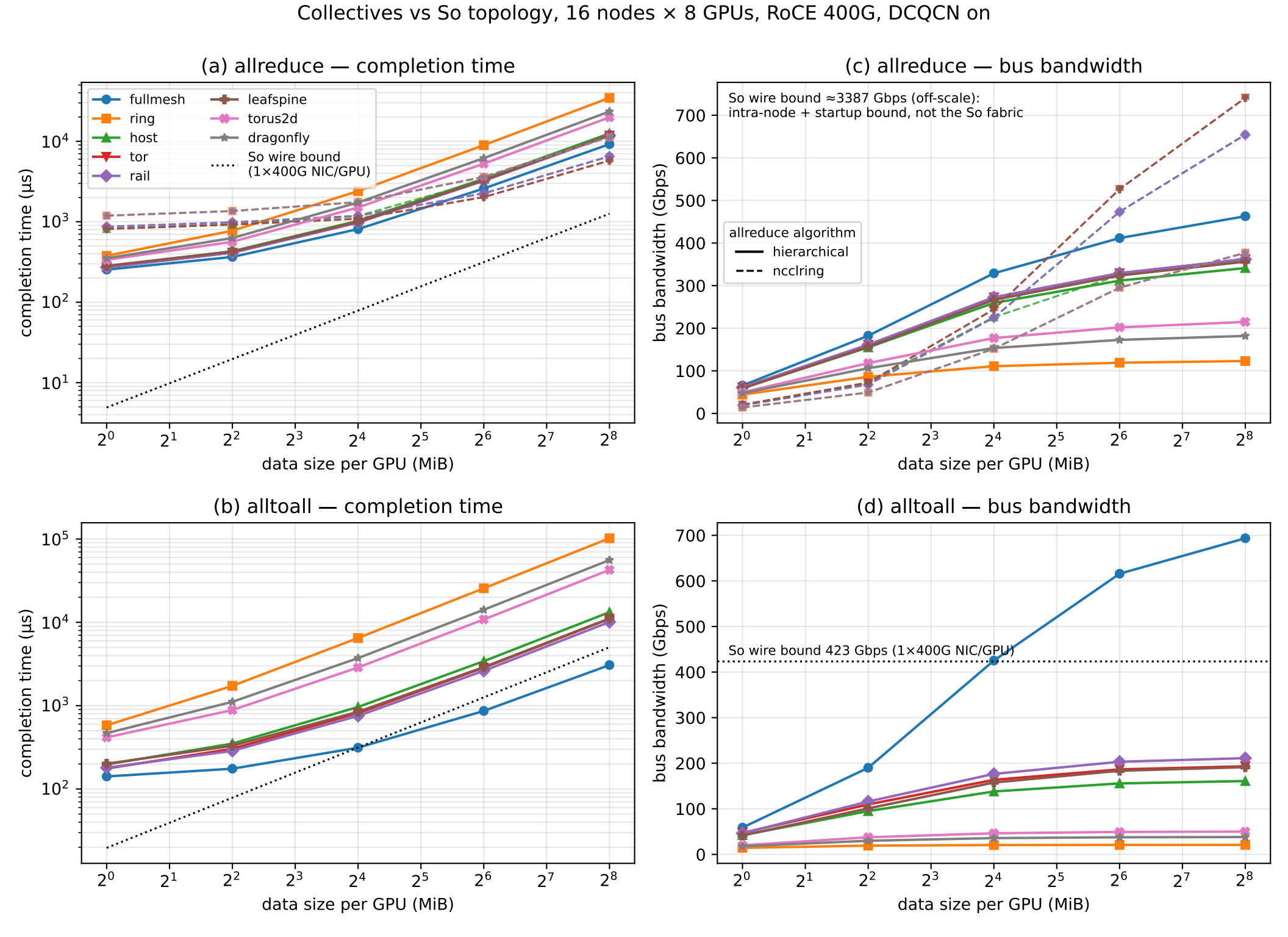
Collectives vs So topology, 16 nodes × 8 GPUs, RoCE 400G, DCQCN on
(a) allreduce — completion time
(b) alltoall — completion time
(c) allreduce — bus bandwidth
(d) alltoall — bus bandwidth
completion time (µs)
bus bandwidth (Gbps)
data size per GPU (MiB)
fullmesh
ring
host
tor
rail
leafspine
torus2d
dragonfly
So wire bound (1×400G NIC/GPU)
So wire bound ≈3387 Gbps (off-scale): intra-node + startup bound, not the So fabric
allreduce algorithm
hierarchical
ncclring
So wire bound 423 Gbps (1×400G NIC/GPU)